

# Agent Applications: Healthcare



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# Agenda

## 1. Deep Research: A Survey of Autonomous Research Agents

(Presented by Tammy Ngo)

## 2. A Survey of LLM-based Agents in Medicine: How far are we from Baymax?

(Presented by Sebastian Pop)

## 3. Evaluating large language models and agents in healthcare: key challenges in clinical applications

(Presented by Matt Juntima)

Presented by Tammy Ngo

**Deep Research: A Survey of Autonomous Research Agents**

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(2025)

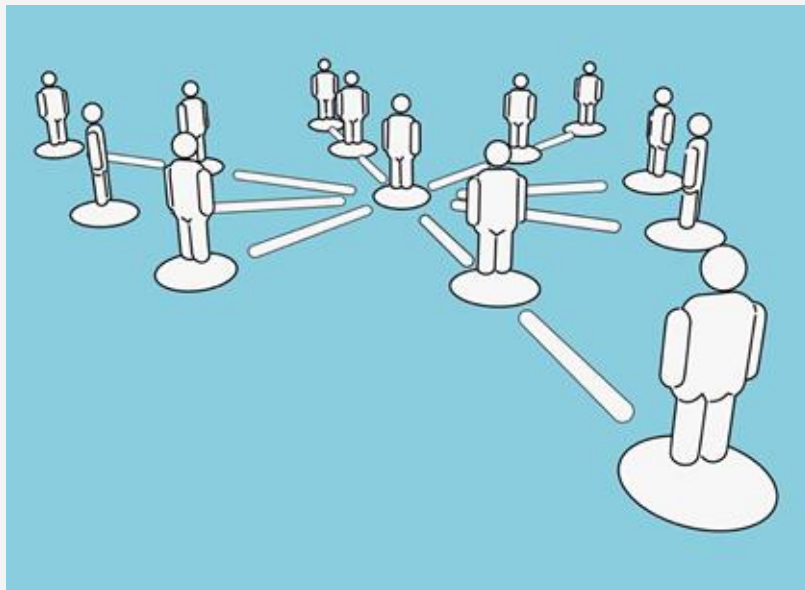
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# Deep Research: A Survey of Autonomous Research Agents

# Motivation

- Large Language Models (LLMs) are powerful but limited by internal knowledge boundaries.
- Autonomous research agents extend LLMs with planning, retrieval, and synthesis abilities.

Goal: Produce **comprehensive, evidence-grounded analytical reports**.

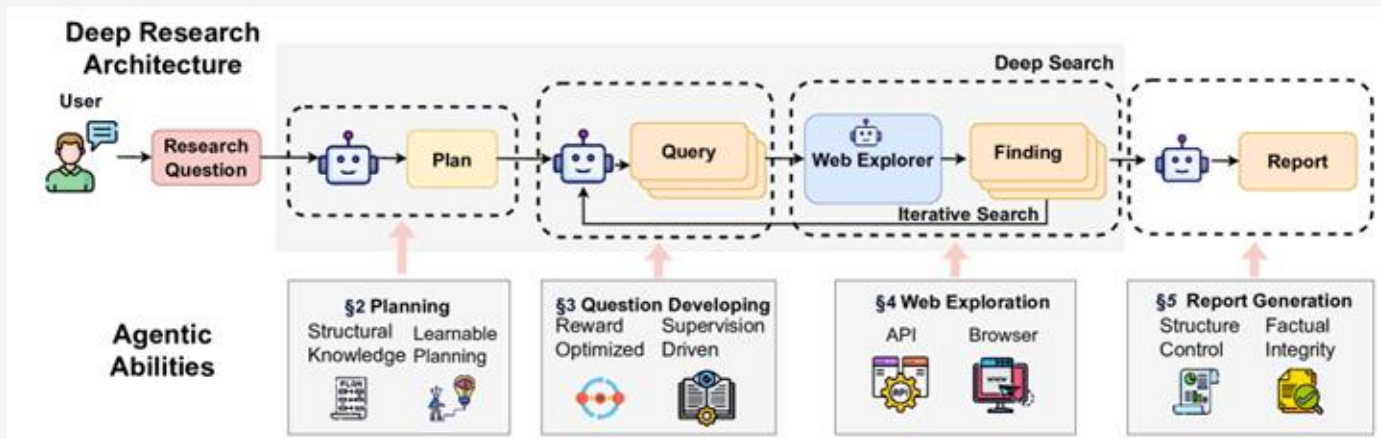


# Deep Research Pipeline

## Four Core Stages

1. Planning
2. Question Developing
3. Web Exploration
4. Report Generation

These stages form the standard pipeline used by modern research agents.





# Research Agent Foundations

# Research Agent Architecture

## Core Modular Components

- Planner / Orchestrator
- Tool & Retrieval Systems
- Reasoning and Synthesis Module
- Memory (short-term + long-term)
- Evaluation and Verification

## Closed-Loop Workflow

**Goal → Plan → Retrieve → Analyze → Evaluate → Refine**

Modular pipelines improve long-horizon task execution and reliability.

# World Models in Research Agents

## LLMs as Implicit World Models

- LLMs encode latent knowledge about tasks, tools, and reasoning patterns.
- Used to predict outcomes of actions during planning.



## Graph-Based Structured Knowledge

- Knowledge graphs provide explicit structure for entities and relationships.
- Improve reasoning traceability and multi-step decision making.
- Combining implicit (LLM) and explicit (graph) world models improves planning reliability.

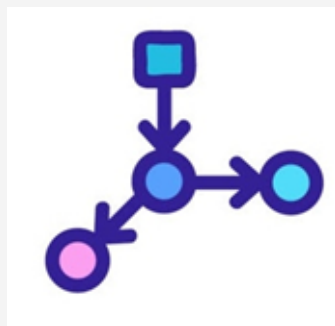




Table 1. Taxonomy of planning methods in deep research agents.

| Planning                                 | Category                         | Related Works                                                                                                                                                                                                                                                 |
|------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Planning with Structured World Knowledge | Planning via simulation          | WebDreameer (Gu et al., <a href="#">2024</a> ), Simulate Before Act (Gu et al., <a href="#">2025</a> )                                                                                                                                                        |
|                                          | Planning via modularity          | Webpilot Zhang et al. ( <a href="#">2025b</a> ), WKM (Qiao et al., <a href="#">2024</a> ), Plan-and-Act Erdogan et al. ( <a href="#">2025</a> )                                                                                                               |
|                                          | Planning via adaptation          | Thought of Search (Katz et al., <a href="#">2024</a> ), MPO (Xiong et al., <a href="#">2025</a> )                                                                                                                                                             |
| Planning as a Learnable Process          | Planning via space exploration   | <b>Agentsquare</b> (Shang et al., <a href="#">2024</a> ), Agent-E Abuelsaad et al. ( <a href="#">2024</a> )                                                                                                                                                   |
|                                          | Planning via self-training       | Patel et al. ( <a href="#">2024</a> ), InSTA Trabucco et al. ( <a href="#">2025</a> )                                                                                                                                                                         |
|                                          | Planning via preference modeling | MindSearch Chen et al. ( <a href="#">2024</a> ), SimpleDeepSearcher (Sun et al., <a href="#">2025b</a> ), Search-in-the-Chain (Xu et al., <a href="#">2024b</a> ), WEPO (Liu et al., <a href="#">2025</a> ), <b>MPO</b> (Xiong et al., <a href="#">2025</a> ) |

# Planning

# Overview & Methods



## Overview

- Planning decomposes research goals into actionable sub-tasks.
- Enables coordination of tools, retrieval, and reasoning.
- Critical for long-horizon and multi-step research workflows.

## Methods

### Planning with Structured World Knowledge

- *Planning via Simulation*: Agents simulate candidate action sequences before execution.
- *Planning via Modularity*: Multi-module or multi-agent pipelines distribute specialized tasks.
- *Planning via Adaptation*: Plans dynamically adjust based on intermediate results and context.

### Planning as a Learnable Process

- *Planning via Space Exploration*: Agents search across possible reasoning or action trajectories.
- *Planning via Self-Training*: Systems iteratively refine their own planning strategies.
- *Planning via Preference Modeling*: Plans optimized toward human-aligned or task-specific objectives.

# Significant Related Works

## Meta-Plan Optimization (MPO)

- Learns how to improve planning strategies across environments.
- Enables adaptive tuning of agent behavior for new tasks.
- Supports transfer of planning knowledge between domains.
- Moves planning from static heuristics to **learned, generalizable policies**.



## AgentSquare

- Automatically assembles agent pipelines from modular components.
- Searches over combinations of planners, tools, memory, and evaluators.
- Identifies efficient architectures for specific research tasks.
- Reduces manual engineering of complex agent workflows.



Table 2. Taxonomy of question developing methods in deep research agents.

| Optimization               | Category                        | Related Works                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reward-Optimized Methods   | Rewards for format and accuracy | DEEPRESEARCHER (Zheng et al., <a href="#">2025</a> ), EVOLVESHARCH (Zhang et al., <a href="#">2025d</a> ), R1-SEARCHER (Song et al., <a href="#">2025</a> ), SEARCH-R1 (Jin et al., <a href="#">2025</a> ), ZEROSEARCH (Sun et al., <a href="#">2025a</a> ), MASKSEARCH (Wu et al., <a href="#">2025b</a> ), DEEPRETRIEVAL (Jiang et al., <a href="#">2025</a> ) |
|                            | Multi-dimensional reward        | INFORAGE (Qian and Liu, <a href="#">2025</a> ), OTC-PO (Wang et al., <a href="#">2025b</a> ), IKEA (Huang et al., <a href="#">2025c</a> ), AUTOREFINE (Shi et al., <a href="#">2025</a> ), R-SEARCH (Zhao et al., <a href="#">2025</a> ), MMSEARCH-R1 (Wu et al., <a href="#">2025a</a> ), VRAG-RL (Wang et al., <a href="#">2025a</a> )                         |
| Supervision-Driven Methods | Multi-agent systems             | MANUSEARCH (Huang et al., <a href="#">2025b</a> ), SEARCH-O1 (Li et al., <a href="#">2025a</a> ), SEARCHAGENT-X (Yang et al., <a href="#">2025b</a> )                                                                                                                                                                                                            |
|                            | Supervision optimization        | REASONRAG (Zhang et al., <a href="#">2025a</a> )                                                                                                                                                                                                                                                                                                                 |

# Question Developing

# Overview & Methods

## Overview

- Converts broad goals into structured research questions.
- Improves retrieval coverage and reasoning depth.
- Essential for complex analytical reporting.

## Methods

### Reward-Optimized Methods

- *Rewards for Format and Accuracy:*
  - Optimize for structured outputs, citations, and evidence grounding.
  - Encourages well-organized and verifiable research questions.
- *Multi-Dimensional Reward:* Joint optimization across relevance, completeness, and reasoning quality.

### Supervision-Driven Models

- *Multi-Agent Systems:* Separate agents generate, critique, and refine questions.
- *Supervision Optimization:* Uses human or synthetic feedback to improve question quality.



Table 3. Taxonomy of web exploration methods in deep research agents.

| Information Source | Category                       | Related Works                                                                                                                                                                                                                                                                                      |
|--------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Web-based          | Web scraping and crawling      | Scrapy (Scrapy developers, <a href="#">2025</a> ), BeautifulSoup (Richardson, <a href="#">2025</a> )                                                                                                                                                                                               |
|                    | Browser-based web agents       | WebGPT (Nakano et al., <a href="#">2021</a> ), Selenium (Selenium contributors, <a href="#">2025</a> )                                                                                                                                                                                             |
|                    | Multimodal web agents          | WebVoyager (He et al., <a href="#">2024</a> ), MM-ReAct (Yang et al., <a href="#">2023</a> ), WebArena (Zhou et al., <a href="#">2023a</a> )                                                                                                                                                       |
| API-based          | Industrial search engines      | Bing (bin, <a href="#">[n.d.]</a> ), X posts (xpo, <a href="#">[n.d.]</a> ), Google (goo, <a href="#">[n.d.]</a> )                                                                                                                                                                                 |
|                    | Domain-specific search engines | Reportify (Reportify, <a href="#">nd</a> ), YanXueZhiDe (China National Knowledge Infrastructure, <a href="#">2024</a> ), CNKI (CNKI, <a href="#">nd</a> ), DuckSearch (Sourty, <a href="#">2024</a> ), BraveSearch (Brave Software, <a href="#">2022</a> ), Bocha (Bocha AI, <a href="#">nd</a> ) |

# Web Exploration

# Overview & Information Sources

## Overview

- Agents gather external evidence through iterative retrieval.
- Extends knowledge beyond model training data.

## Information Sources

### Web-Based Exploration

- *Web Scraping and Crawling*: Automated extraction of documents and structured information.
- *Browser-Based Web Agents*: Navigate pages, follow links, and interact with interfaces.
- *Multimodal Web Agents*: Process text, images, tables, and mixed media sources.

### API-Based Exploration

- *Industrial Search Engines*: Provide scalable, reliable web retrieval.
- *Domain-Specific Search Engines*: Access specialized databases, such as academic or technical sources.



Table 4. Taxonomy of artifact generation methods in deep research agents.

| Field             | Category                     | Representative Works                                                                                                                                                            |
|-------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Structure Control | Planning-based Generation    | Agent Laboratory (Schmidgall et al., 2025), AI Scientist v2 (Yamada et al., 2025), LongEval (Wu et al., 2025c), LongWriter (Bai et al., 2024), LongDPO (Ping et al., 2025)      |
|                   | Constraint-guided Generation | WebThinker (Li et al., 2025b), Suri (Pham et al., 2024), Wan et al. (2025)                                                                                                      |
|                   | Structural-aware Evaluation  | Long2RAG (Qi et al., 2024), ExPerT (Salemi et al., 2025), Long et al. (2025), Kim et al. (n.d.), Huang et al. (2024)                                                            |
| Factual Integrity | Faithful Modeling            | RAGSynth (Shen et al., 2025), BRIDGE (Dai et al., 2025), Zhou et al. (2023b), Shi et al. (2024)                                                                                 |
|                   | Conflict Reasoning           | FaithfulRAG (Zhang et al., 2025c), DRAGged (Cattan et al., 2025), Yuan et al. (2024), Ying et al. (2023)                                                                        |
|                   | Factuality Evaluation        | Face4RAG (Xu et al., 2024a), SFR-RAG (Nguyen et al., 2024), Wallat et al. (2024), FaithJudge (Tamber et al., 2025), RAG-QA Arena (Han et al., 2024), MT-RAIG (Seo et al., 2025) |

# Report Generation



# Overview & Methods

## Overview

- Synthesizes retrieved evidence into structured analytical reports.
- Emphasizes traceability and citation support.

## Methods

### Structure Control

- *Planning-Based Generation*: Uses hierarchical outlines to guide report structure.
- *Constraint-Guided Generation*: Enforces formatting, section coverage, and citation rules.
- *Structural-Aware Evaluation*: Measures organization, coherence, and completeness.

### Factual Integrity

- *Faithful Modeling*: Grounds outputs in retrieved evidence.
- *Conflict Reasoning*: Detects and reconciles contradictory sources.
- *Factuality Evaluation*: Measures evidence support and citation correctness.



# Optimization



# Optimization Techniques

- Prompt optimization.
- Supervised fine-tuning.
- Reinforcement learning for agent policies.
- Multi-objective optimization across accuracy, coverage, and efficiency.
- Optimization improves stability and long-horizon performance.



Table 5. Coverage of core Deep Research modules across major benchmarks. P: Planning, QD: Question Developing, WE: Web Exploration, RG: report generation.

| Benchmark                                | P | QD | WE | RG | Task                       | Evaluation Metrics                                   |
|------------------------------------------|---|----|----|----|----------------------------|------------------------------------------------------|
| MIND2WEB 2 (Gou et al., 2025)            | ✓ | ✓  | ✓  | X  | Web search                 | Success rate, Partial Completion                     |
| BROWSECOMP (Wei et al., 2025)            | ✓ | ✓  | ✓  | X  | Web search                 | Accuracy, Calibration Error                          |
| WEBARENA (Zhou et al., 2023a)            | ✓ | ✓  | ✓  | X  | Web search                 | Success Rate                                         |
| GAIA (Mialon et al., 2023)               | ✓ | ✓  | ✓  | X  | Multi-step assistant tasks | EM                                                   |
| HUMANITY'S LAST EXAM (Phan et al., 2025) | ✓ | ✓  | ✓  | X  | Multidomain reasoning      | Accuracy, Calibration Error                          |
| BROWSECOMP-ZH (Zhou et al., 2025)        | ✓ | ✓  | ✓  | X  | Web search in Chinese      | Accuracy, Calibration Error                          |
| MEDBROWSECOMP (Chen et al., 2025b)       | ✓ | ✓  | ✓  | X  | Medical web search         | Accuracy                                             |
| GPQA (Rein et al., 2024)                 | ✓ | ✓  | ✓  | X  | Graduate-level QA          | Accuracy                                             |
| INFODEESEEK (Xi et al., 2025)            | X | ✓  | ✓  | X  | Open-domain QA             | Accuracy, Information Accuracy                       |
| DEEPRESEARCH BENCH (Du et al., 2025)     | ✓ | ✓  | ✓  | ✓  | Research report generation | Pairwise Agreement Rate, Overall Pearson Correlation |
| DEEPRESEARCHGYM (Coelho et al., 2025)    | ✓ | ✓  | ✓  | ✓  | Research task sandbox      | KPR, KPC, Precision, Recall, Clarity, Insight        |

# Benchmarks and Evaluation

# Overview & Benchmarks

## Overview

- Research agents require evaluation beyond simple QA accuracy.
- Metrics target reasoning quality and report usefulness.

## Benchmarks

### Search-Oriented Benchmarks

- Evaluate retrieval effectiveness and navigation ability.
- Measure tool use and exploration strategies.

### Research-Oriented Benchmarks

- *DeepResearch Bench*
  - Evaluates:
    - Report fidelity
    - Citation accuracy
    - Comprehensive topic coverage
  - Uses complex, multi-step research tasks.





# Limitations and Future Directions

# Key Challenges

## Plan Brittleness

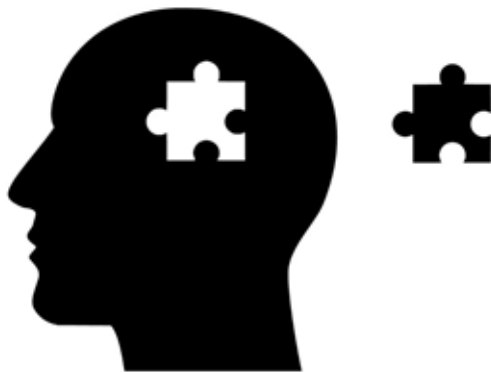
- Plans may fail under uncertainty or unexpected results.

## Lack of Robustness to Ambiguous Queries

- Agents struggle when goals are underspecified.

## Evaluation Coarseness

- Existing metrics do not fully capture research quality.



# Future Directions

## Multi-Tool Integration

- Better coordination across search, code, and analysis tools.

## Factuality Improvements

- Stronger verification and grounding mechanisms.

## Multimodal Reasoning Capabilities

- Integrating text, images, tables, and data.

## Workflow Design and Model Optimization

- More adaptive and efficient agent pipelines.

## Personalization

- Tailoring research outputs to user intent and expertise.





# Key Takeaways

- Autonomous research agents shift AI from **single responses** to **iterative, evidence-grounded research workflows**.
- Future progress depends on:
  - Robust planning
  - Better evaluation
  - Reliable factual reasoning





## **A Survey of LLM-based Agents in Medicine: How far are we from Baymax?**

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# A Survey of LLM-based Agents in Medicine: How far are we from Baymax?

# Motivation

## Context

- Explosion of LLMs (ChatGPT, GPT-4, etc) → hype in healthcare
- Traditional LLMs  $\neq$  enough for real medical workflows

## Medical workflows require more than good answers

- Structured reasoning
- Decision making
- Tool usage

## Toward LLM-based agents

- Need for LLM-based agents
- Combining language models with planning, memory, and external tools → support medical tasks

*The “Baymax” vision: safe, autonomous medical assistants*



# Background

## **Scope**

- Reviews 60 studies on LLM-based medical agents (2022-2024)

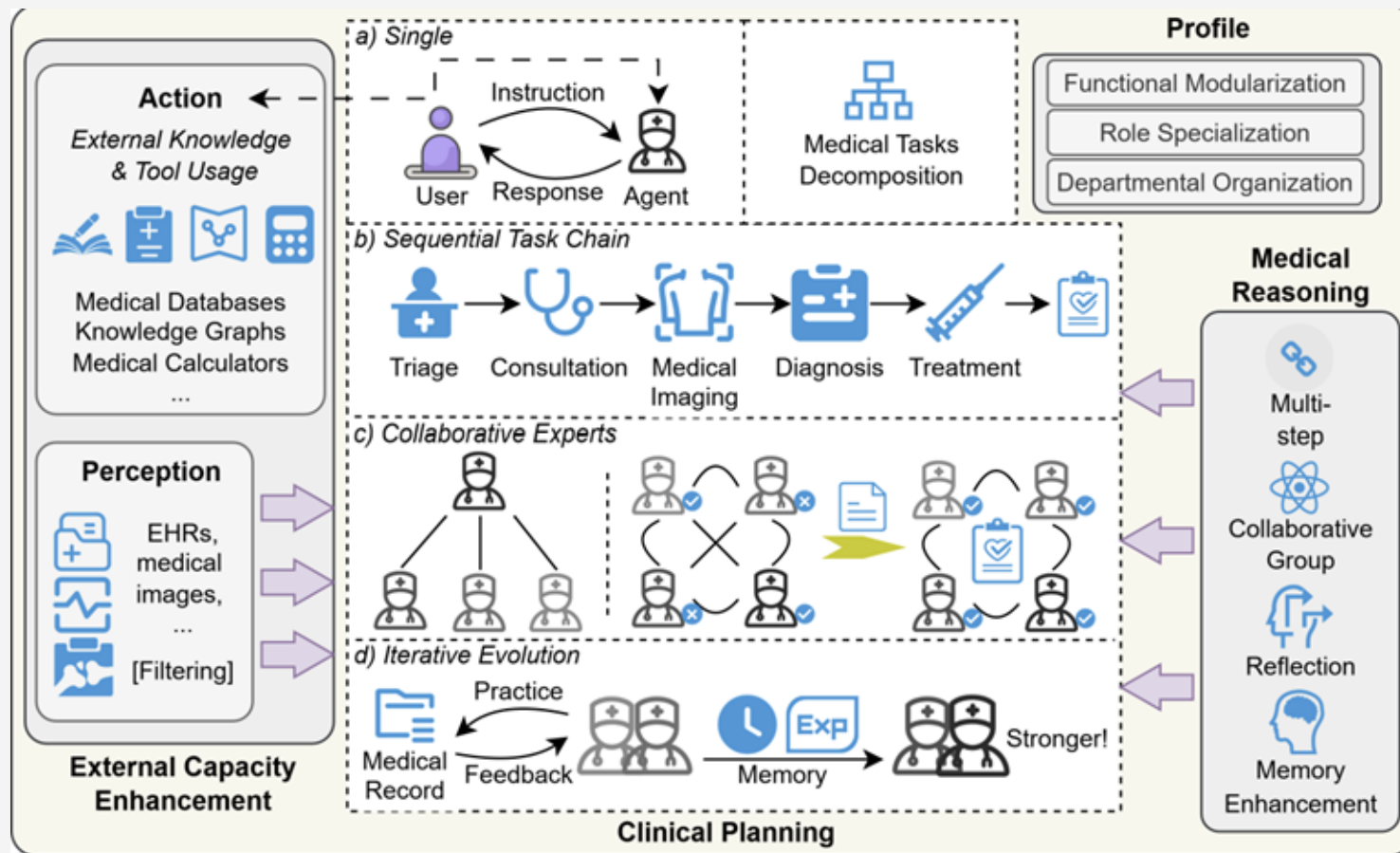
## **Rather than focusing on individual models**

- Examines agent-based systems embedded in structured workflows

## **Medical settings**

- These agents must operate under stricter constraints than general AI systems
  - Including handling multimodal data (clinical text, medical imaging, and lab results)

# Architecture Overview



# Agent Profiles

## What is an agent profile?

- Define roles and responsibilities of an LLM-based medical agent

## How profiles are assigned

- Rather than a single generic agent, many systems assign specific profiles
- Reflects how work is organized in real clinical settings

## Three common approaches

- Profiles based on functional modules (e.g., data analysis)
- Profiles that mirror medical roles (e.g., specialists)
- Profiles organized by medical departments (e.g., radiology)

## Why this matters

- Defining clear profiles allow agents to specialize
- Coordinate with each other
- Collaborate more effectively with clinicians in complex workflows



# Clinical Planning & Agent Paradigms

## What is clinical planning?

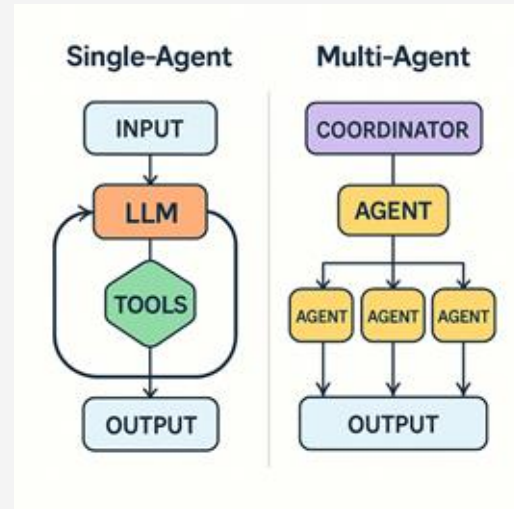
- Describes how LLM-based medical agents structure and coordinate complex medical workflows

## Rather than a single response, agents break work into structured steps:

- Data collection → Diagnosis → Treatment planning → follow-up

## Identifies several common planning paradigms

- In simpler settings → single agent may handle the entire task
- More complex workflows:
  - Sequential task chains (steps handled in order)
  - Collaborative expert agents (specialized agents work in parallel)
- Enables medical agents to operate
  - More reliable, transparent, and controllable way than standalone language models



# Medical Reasoning

## What is medical reasoning?

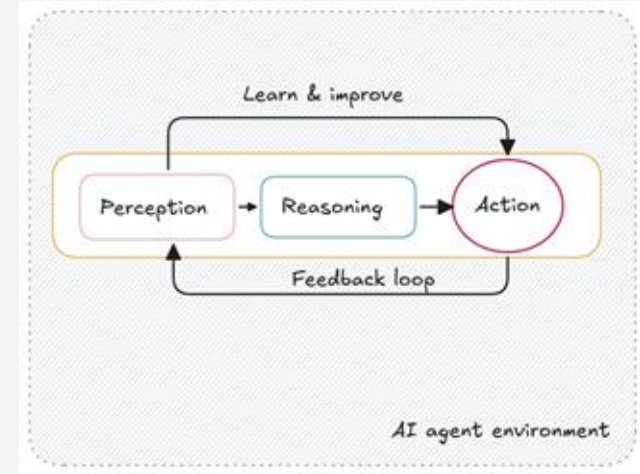
- Describes how LLM-based agents generate, evaluate, and refine clinical decisions

## Unlike traditional LLMs that produce one-shot answers

- Medical agents often rely on multi-step reasoning processes
- Break down complex cases into intermediate steps

## Reflection & collaboration

- Agents review their own outputs
- Receive feedback from other agents → correct errors & reduce uncertainty





# External Capacity Enhancement

**Focus on how LLM-based medical agents connects to real-world clinical resources**

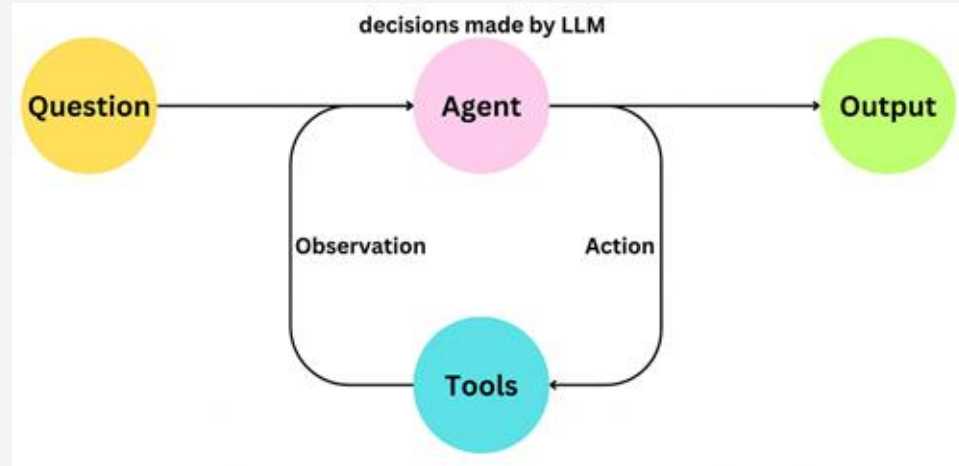
- Rather than relying only on internal knowledge

**Agents access external data and tools**

- Electronic health records
- Medical images
- Clinical guidelines
- Drug databases
- Use specialized tools:
  - Medical calculators
  - Knowledge graphs
- Support more accurate and context aware decisions

**External grounding is important in healthcare**

- Up-to-date information
- Verifiable sources are essential for safe deployment



# Application Scenarios

- **Clinical decision support and diagnosis**
- **Clinical data analysis and documentation**
- **Medical training and simulation**
- **Healthcare service optimization**

# Clinical Decision Support & Diagnosis

## Agents mainly used to improve reasoning quality

- Rather than to replace clinicians

## How agents improve decisions

- Many systems have agents with specialized roles (diagnosis, evaluation, treatment)
- Collaboration helps reduce bias and improve accuracy

## Applied in areas

- Diagnosis, prescription checking, and patient interaction safety



# Other Application Areas

## **Clinical data analysis & documentation**

- *Summarizing records, predicting outcomes, generating patient-friendly reports*

## **Medical training**

- *Agent-based simulations to evaluate and improve diagnostic and treatment skills in realistic environments*

## **Healthcare service optimization**

- *Automating non-diagnostic tasks (patient education and administration), helping reduce clinicians workload while maintaining service quality*

| Purpose                                  | Functionality                                 |                                   | Work                     | Framework Type        | Tool Use |
|------------------------------------------|-----------------------------------------------|-----------------------------------|--------------------------|-----------------------|----------|
| Clinical Decision Support and Diagnosis  | Refine Diagnostic Reasoning                   |                                   | (Dutta and Hsiao, 2024)  | Adaptive Planning     | -        |
|                                          | Reduce Cognitive Bias                         |                                   | (Ke et al., 2024)        | Collaborative Experts | -        |
|                                          | Task Coordination                             |                                   | (Wei et al., 2024b)      | Sequential Task Chain | -        |
|                                          | Diagnosis Accuracy                            |                                   | (Kim et al., 2024c)      | Collaborative Experts | Yes      |
|                                          |                                               |                                   | (Tang et al., 2024)      | Collaborative Experts | -        |
|                                          | Domain Specific Functionalities               | Clinical Trial Outcome Prediction | (Yue et al., 2024)       | Sequential Task Chain | Yes      |
|                                          |                                               | Patient Interaction Safety        | (Mukherjee et al., 2024) | Sequential Task Chain | -        |
|                                          |                                               | Prescription Validation           | (Van et al., 2024)       | Sequential Task Chain | Yes      |
|                                          | Diagnosis Capability                          |                                   | (Yan et al., 2024)       | Collaborative Experts | -        |
| Integrated Modelling                     |                                               | (Fan et al., 2024)                | –                        | -                     |          |
| Clinical Data Analysis and Documentation | Mortality Prediction                          |                                   | (Wang et al., 2024b)     | Collaborative Experts | Yes      |
|                                          | Hospital Readmission Analysis                 |                                   |                          |                       |          |
|                                          | Clinical Documentation                        |                                   | (Lee et al., 2024)       | Single Agent          | -        |
|                                          | Patient Friendly Medical Reports              |                                   | (Sudarshan et al., 2024) | Iterative Evolution   | Yes      |
|                                          | Integrated Simulation                         |                                   | (Li et al., 2024b)       | Iterative Evolution   | -        |
| Medical Training and Simulation          | Evaluated Diagnosis and Treatment Performance |                                   | (Yan et al., 2024)       | Collaborative Experts | -        |
|                                          | Integrated Simulation                         |                                   | (Fan et al., 2024)       | –                     | -        |
|                                          |                                               |                                   | (Li et al., 2024b)       | Iterative Evolution   | -        |
|                                          | Medical Training                              | Training Environment              | (Wei et al., 2024a)      | Collaborative Experts | Yes      |
|                                          |                                               |                                   | (Wu et al., 2024)        | Collaborative Experts | -        |
|                                          |                                               | Scenario Simulation               | (Yu et al., 2024)        | Collaborative Experts | Yes      |
| Healthcare Service Optimization          | Automation of Non-diagnostic Tasks            |                                   | (Mukherjee et al., 2024) | Sequential Task Chain | Yes      |
|                                          |                                               |                                   | (Laymouna et al., 2024)  | –                     | -        |
|                                          | Automation of Diagnostic Tasks                |                                   | (Chadebecq et al., 2023) | Iterative Evolution   | -        |

# Evaluation & Benchmarking

## Why evaluation is critical

- Medical agents operate in high-risk clinical environments, where errors can directly affect patient safety
- Careful evaluation is required to ensure reliability, safety, and clinical usefulness

## Benchmark categories:

- Static Q&A benchmarks
- Workflow-based simulations
- Automated evaluation frameworks

## Shows that different evaluation methods capture different aspects of medical AI agent performance

- Meaning that no single benchmark is sufficient on its own

| Evaluation Attribute                 | Genre                                | Specific Names                           | Related Work                                           |
|--------------------------------------|--------------------------------------|------------------------------------------|--------------------------------------------------------|
| Benchmarks                           | Static Q&A Benchmarks                | <i>MedQA</i>                             | (Jin et al., 2020)                                     |
|                                      |                                      | <i>MedMCQA</i>                           | (Pal et al., 2022)                                     |
|                                      |                                      | <i>Pub-MedQA</i>                         | (Jin et al., 2019)                                     |
|                                      |                                      | <i>MMLU</i>                              | (Hendrycks et al., 2021b)<br>(Hendrycks et al., 2021a) |
|                                      | Workflow-Based Simulation Benchmarks | <i>MedChain</i>                          | (Liu et al., 2024a)                                    |
|                                      |                                      | <i>AI Hospital</i>                       | (Fan et al., 2024)                                     |
|                                      |                                      | <i>AgentClinic</i>                       | (Schmidgall et al., 2024b)                             |
|                                      |                                      | <i>ClinicalLab</i>                       | (Yan et al., 2024)                                     |
|                                      | Automated Evaluation Frameworks      | <i>AI-SCE</i>                            | (Mehandru et al., 2024)                                |
|                                      |                                      | <i>RJUA-SPs</i>                          | (Liu et al., 2024b)                                    |
| Metrics for Task-Specific Evaluation | Exact Match Metrics                  | Accuracy                                 | –                                                      |
|                                      |                                      | Precision                                | –                                                      |
|                                      |                                      | Recall                                   | –                                                      |
|                                      | Semantic Similarity Metrics          | <b>BLEU</b>                              | (Papineni et al., 2002)                                |
|                                      |                                      | <b>ROUGE</b>                             | (Lin, 2004)                                            |
|                                      |                                      | <b>BertScore</b>                         | (Zhang et al., 2020)                                   |
|                                      | LLM-Based Evaluation Metrics         | <i>ChatCoach</i>                         | (Huang et al., 2024a)                                  |
|                                      |                                      | Retrieval-Augmented Evaluation Framework | (Liu et al., 2024b)                                    |

# Challenges & Limitation

## Hallucinations

- Incorrect or fabricated outputs are dangerous in medical settings
- Errors in diagnosis or treatment can directly affect patient outcomes

## Evaluation methods

- Many current benchmarks rely on static tasks, and fail to capture real clinical workflows, interactive decision-making, and reasoning quality

## Integrating multimodal data

- Medical agents must combine text, images and laboratory results
- Differences in data formats, documentation standards, and language can make it difficult

## Large-scale deployment

- Raises practical & ethical concerns (high costs, privacy and security tasks, algorithmic bias, and the lack of patient-centered design)

# Future Directions

## **Strong focus on improving medical reasoning**

- Through multi-step inference and reflection

## **Simulation-based training**

- Key way to evaluate diagnosis and treatment decisions more realistically

## **Better integration with clinical systems**

- Physical devices (monitoring systems and medical robotics)

## **Should prioritize safety before autonomy**

- Current LLM-based agents should be viewed as assistive tools rather than autonomous clinicians





# Key takeaways

## **LLM-based medical agents go beyond passive text generation**

- They enable planning, multi-step reasoning, and tool use within clinical workflows

## **Agent architectures are essential for medical AI**

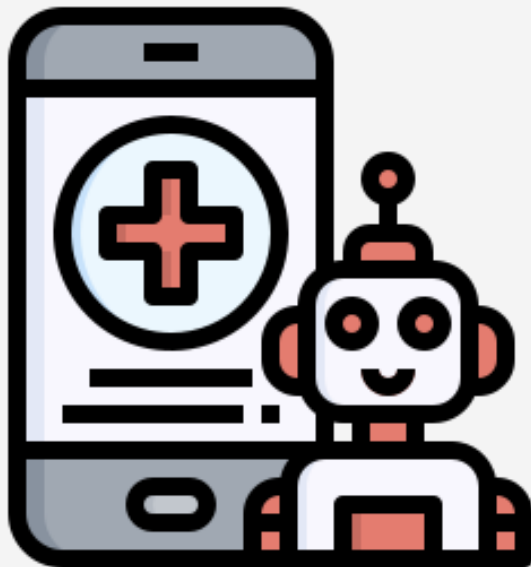
- Profiles, planning, reasoning, and external tools support safe and reliable system

## **Multi-agent and sequential workflows dominate practice**

- Especially in clinical decisions support and diagnostic reasoning

## **We are not yet at Baymax**

- Current systems show promise, but remain safety-focused and clinician-assisted



Review

# Evaluating large language models and agents in healthcare: key challenges in clinical applications

Xiaolan Chen<sup>1 #</sup>, Jiayang Xiang<sup>2 #</sup>, Shanfu Lu<sup>3 #</sup>, Yexin Liu<sup>4</sup>, Mingguang He<sup>1 5 6</sup>  , Danli Shi<sup>1</sup>  
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Evaluating large language models and agents in healthcare: key challenges in clinical applications

# Background/Motivation

- LLMs are very useful in conversations and have been applied to many different industries
- Ideas with integrating LLM agents to the healthcare space
- If LLM agents has hallucinations in healthcare, could be dangerous for anyone involved
- Currently no good way of evaluating if certain LLM agents are safe for medical use
- Patients, doctors, administrators, and many more stakeholders will benefit with better evaluated LLM agents.



## Problem

1. Data bias - datasets from specific domains or populations
2. Evaluations across diverse healthcare applications lack depth - fails to differentiate strengths and weaknesses of LLMs in practical scenarios
3. Prior evaluation methods primarily focus on accuracy, but overlooks critical things hallucinations, logical reasoning, and the likelihood of generating harmful content



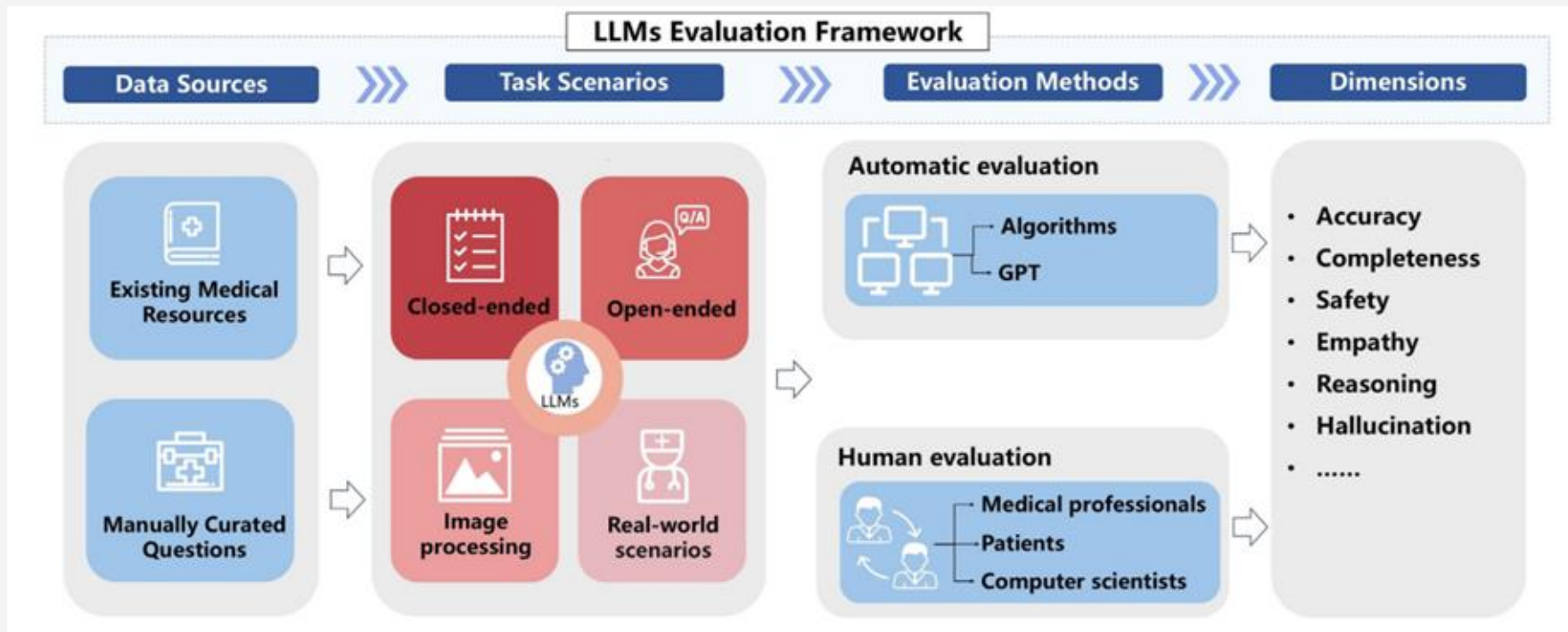
## Solution

1. Expanding of data sources and task scenarios
2. Create diverse evaluation methodologies from objective accuracy metrics to more nuanced human-centric evaluations
3. Identified key challenges and opportunities in medical LLM and LLM agent evaluation

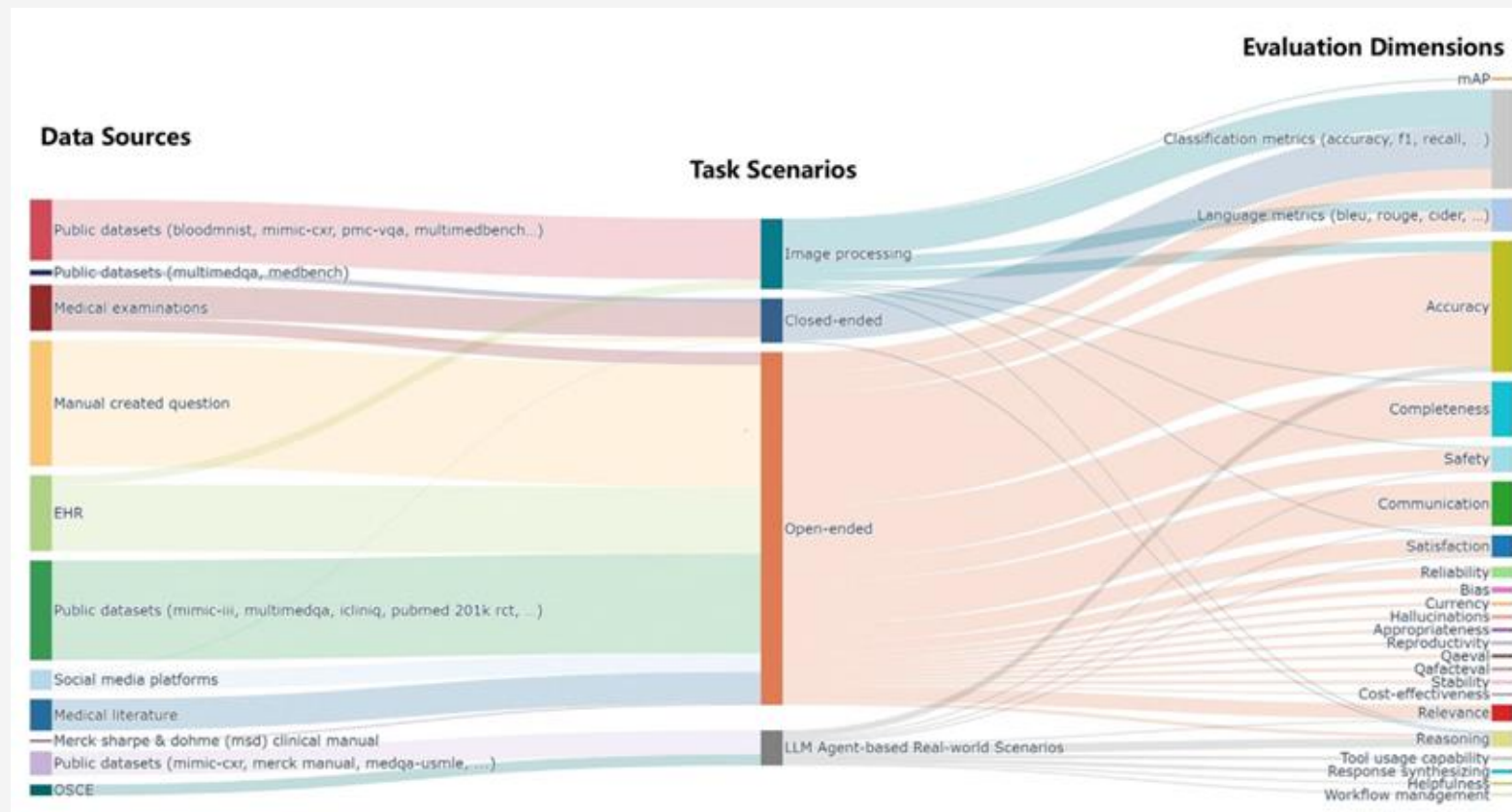
# Experimentation Methodology

- Systematic search was carried out across PubMed, Google Scholar, and Web of Science databases between 2023-2024
- Datasets can be broadly divided into 2 main categories: existing medical resources and manually curated question sets
- Different tasks such as close-ended and open-ended tasks, summarizing, information extraction, etc.

# Solution

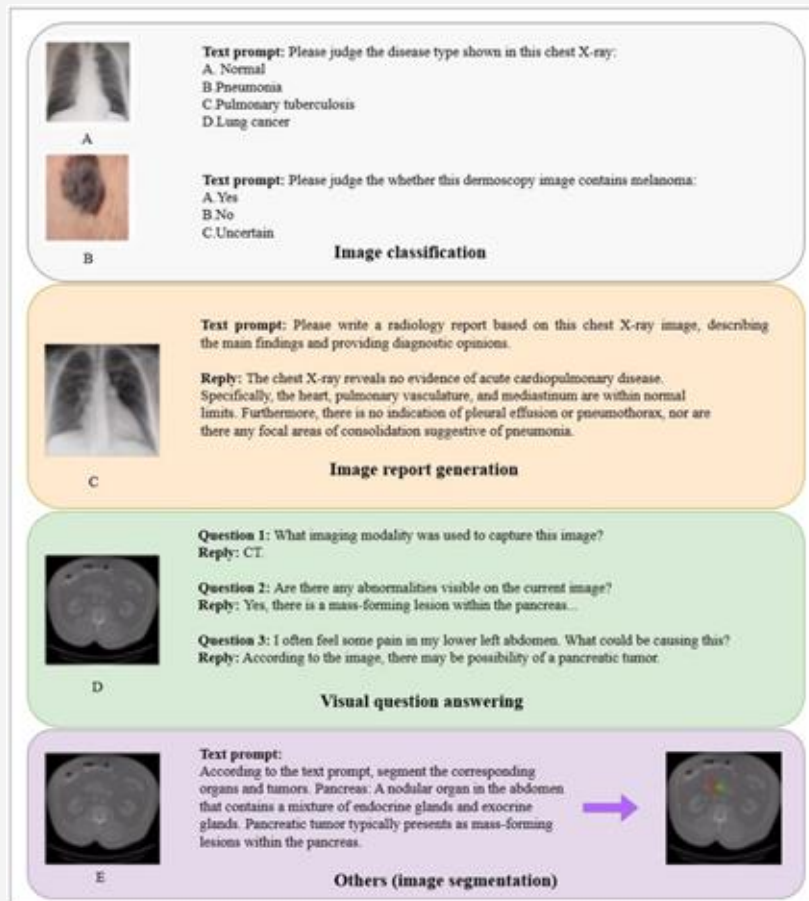


# Summary of Data Sources/Evaluation Dimensions



# Tasks scenarios

- **Close-ended tasks**
- **Open-ended tasks:** Summarization, Information extraction, Medical question answering
- **Image processing tasks:** Image classification, Report generation, Visual question answering, Other scenarios
- **Real-world multitasking involving LLM agents**





# Closed Ended Tasks

- Evaluated models such as GPT-4 on the CMedExam dataset and found that its **61.6%** accuracy was still significantly lower than human level (**71.6%**)
- GPT-3 and GPT-4 evaluated with the American College of Gastroenterology self-assessment tests. A score of **70%** or higher is generally required to pass the evaluation. ChatGPT-3.5 and ChatGPT-4 scored **65.1%** and **62.4%**, respectively.
- Closed ended tasks are easy to test but are not complex

# Open Ended Tasks

Divided into **summarization**, **information extraction**, and **medical QA**:

- **Summarization:**
  - Automatic metrics were used to assess the lexical and semantic similarity, and human evaluation was conducted on coherence, factual consistency, comprehensiveness, and harmfulness
  - LLMs could be susceptible to generating incorrect information and lead to potential harm due to misinformation
  - Novel feasibility of using LLMs to generate patient-friendly summaries of radiology reports

# Open Ended Tasks Continued

## Information Extraction:

- Numerous studies showed LLMs have the potential to assist researchers or patients search and acquire knowledge from a large amount of biomedical data
- ChIP-GPT had a 90%–94% accuracy when trained with 100 examples.

## Medical question answering:

- Performance of ChatGPT - considered average
- Models need to be specifically trained for individual medical subspecialties
- ChatGPT demonstrates empathy when answering patients' questions.

# Image processing tasks

Divided into **image classification**, **report generation**, **visual QA**, and **other scenarios**

## **Image classification:**

- The results showed that LLMs performed well on different medical image tasks but still need improvement in recognizing fine-grained visual concepts
- Results indicated that while CNN models excelled due to specialized training, vision-language models (VLMs) showed promise in zero-shot and few-shot scenarios without prior training

# Image processing tasks

## **Report Generation:**

- Model could generate descriptive reports with structured prompts, but there is still space for improvement in the accuracy of generating specific medical terms

## **Visual QA:**

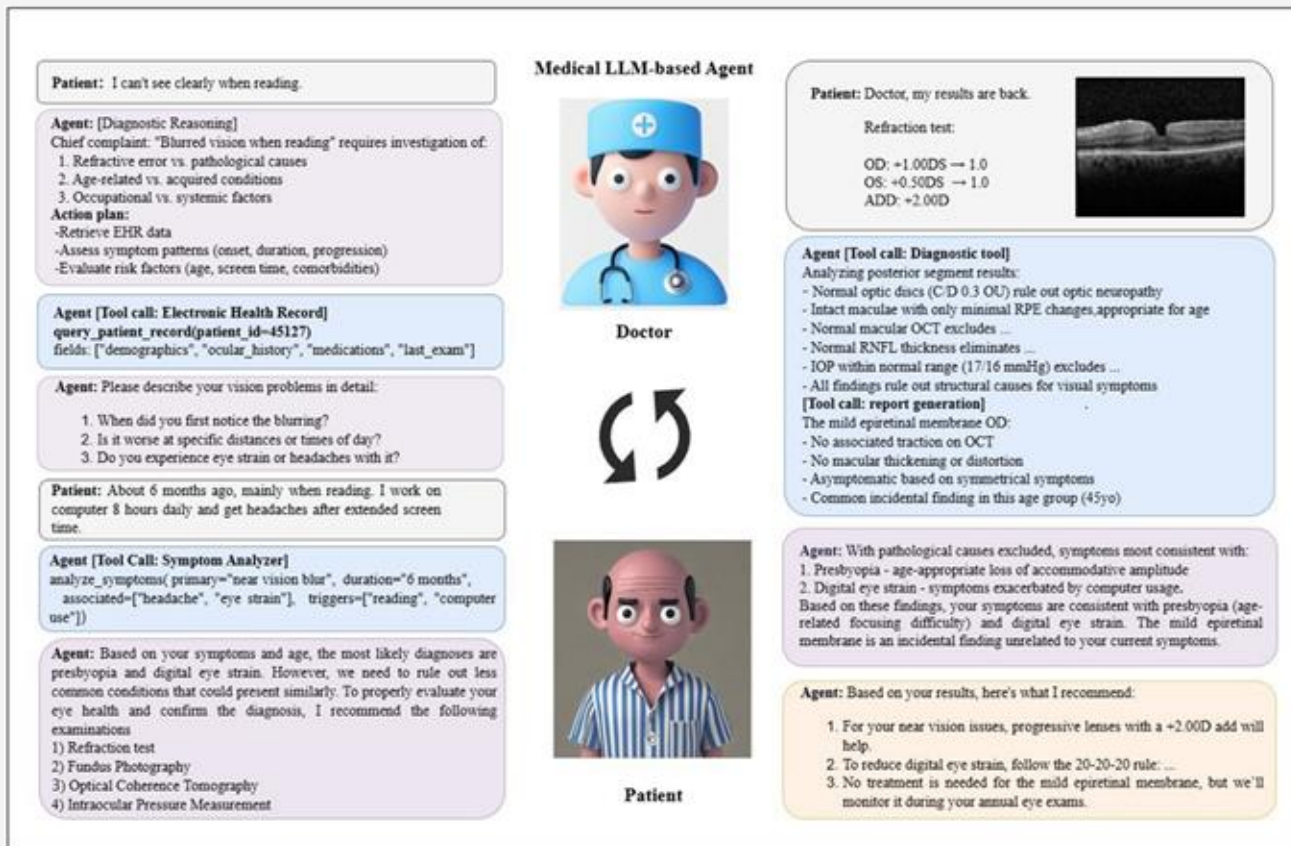
- Requires stronger cross-modal understanding capabilities and higher demands on natural language expression.
- Constructing large scale, high-quality datasets is crucial for evaluating model performance.

# Image processing tasks

## Other Scenarios:

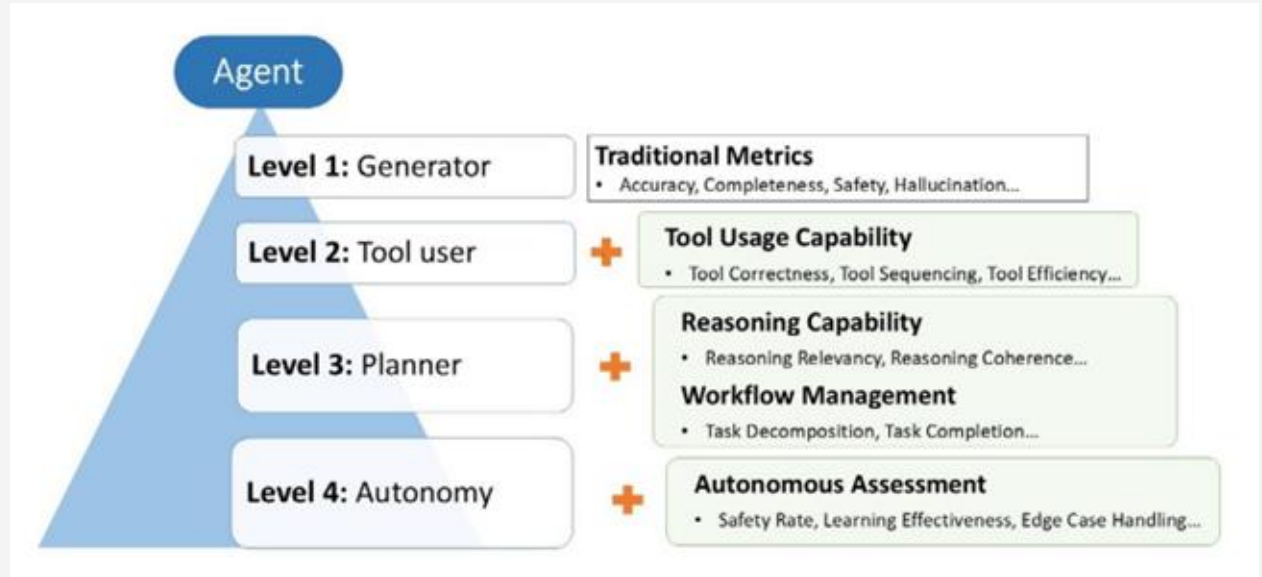
- LLMs also show good prospects in other medical image analysis tasks such as medical image segmentation and cross-modal retrieval
- Current evaluations of LLMs in these new tasks are inadequate

# Real-world Multitask Example



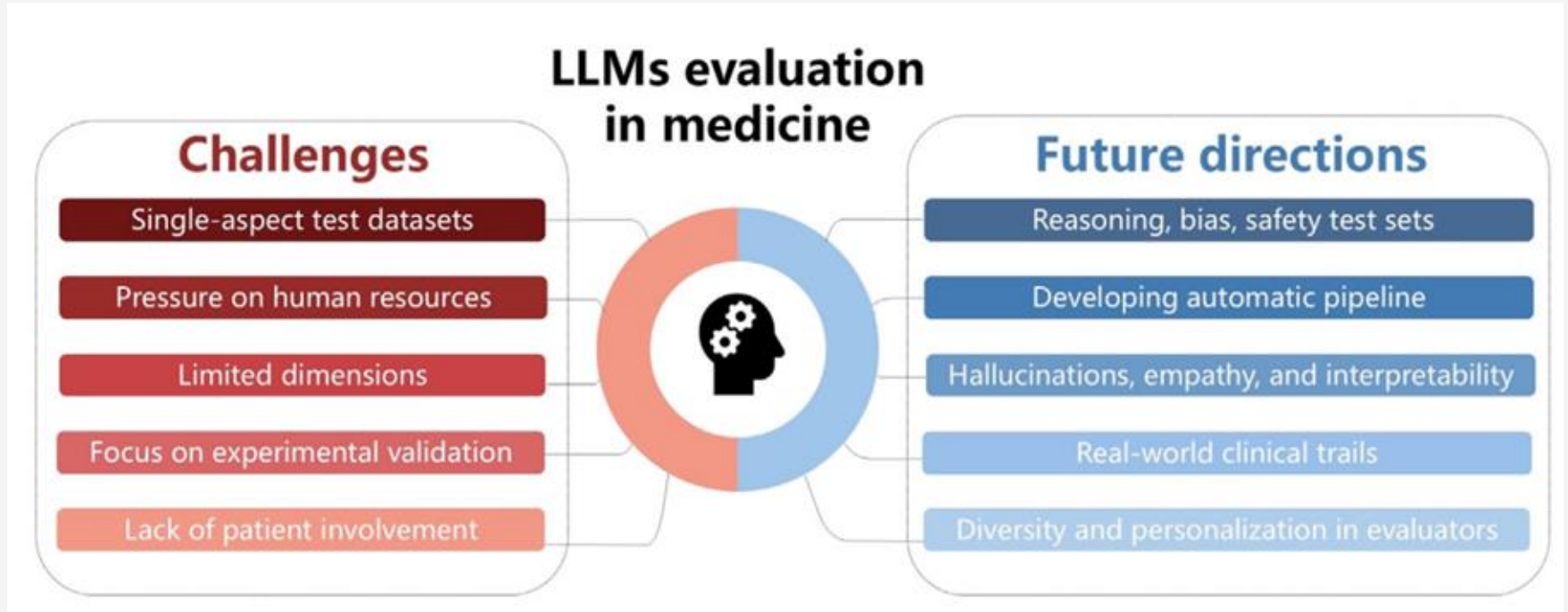
# Evaluation Methodology

- Traditional evaluation dimensions - accuracy
- LLM agent evaluation dimensions





# Challenges/Future Directions



# Contributions

1. Summarized data sources used in evaluations - includes existing medical resources and manually designed clinical questions
2. Analyzed key medical task scenarios
3. Compared evaluation methods and dimensions



# Conclusion

- Paper argues for a precise and effective evaluation framework, current standards are not good enough
- The requirements for a good framework is what follows:
  1. Using high quality datasets
  2. Combining human evaluation with automated evaluation
  3. Setting comprehensive and effective assessment standards beyond accuracy
  4. Including specific LLM agent evaluation methods
  5. Having future trial designs being more aligned with real-world clinics and compare LLMs to existing practices



## Recap

# Paper Summaries

## Paper 1

Modern LLM-based agents autonomously conduct complex research by...

- Integrating planning
- Adaptive query generation
- Iterative web exploration
- Structured report synthesis

to produce comprehensive, evidence-grounded analyses

## Paper 2

LLM-based agents show promise in healthcare, but current systems remain assistive and require structured planning, reasoning and evaluation

## Paper 3

Need for better datasets and more diverse evaluation techniques to evaluate LLMs in healthcare



# Questions